Alena Malyarenko

Alena Malyarenko | alenamalyarenko | Alena Malyarenko

EDUCATION

PhD (Physics)
2020 - Ōtākou Whakaihu Waka | University of Otago, Te Tari Hū-o-te-Kōhao | Department of Physics and Taihoro Nukurangi | The National Institute of Water and Atmospheric Research, Te Whanganui-a-Tara — Wellington

MSc in Hydrometeorology (with Distinction)
2015 - Lomonosov Moscow State University, Department of Oceanography, Faculty of Geography

BSc in Hydrometeorology (with Distinction)
2013 - Lomonosov Moscow State University, Department of Oceanography, Faculty of Geography

EMPLOYMENT

Postdoctoral Researcher
Jul 2023 - present: Te Whare Wānanga o Waitaha | University of Canterbury
Te Kura Aronukurangi | School of Earth and Environment

Adjunct Postdoctoral Researcher
Feb 2020 - present: Te Herenga-Waka | Victoria University of Wellington
Te Puna Pātiotio | Antarctic Research Centre, The National Modelling Hub

Postdoctoral Researcher
Feb 2020 - Jun 2023: Taihoro Nukurangi | The National Institute of Water and Atmospheric Research
Ocean Physics

Researcher
2014 - 2016: Hydrometeorological Centre of Russia, Laboratory of Arctic Hydrometeorology

Researcher
2011 - 2013: Lomonosov Moscow State University, Natural Risk Assessment Laboratory

RESEARCH GRANTS

Current Funding

Program Title:	Antarctic Science Platform Phase Two - Tiaki Moana – What are the drivers and potential implications of unprecedented, climate-driven change in the Ross Sea and Southern Ocean?
Funding Agency:	Hīkina Whakatutuki | Ministry of Business Innovation & Employment, Strategic Science Investment Fund
Role:	Key Researcher
Total Budget:	$49 million
Period:	2025–2032
My Funding:	0.2 FTE per annum, 2027-2032 ($215,000)
My Contribution:	Understanding the oceanic pathways of warm water into ice cavities due to changes in sea ice conditions using the P-SKRIPS model.

Program Title:	Antarctic Sea-Ice Switch - Preparing for New Threats
Funding Agency:	Hīkina Whakatutuki | Ministry of Business, Innovation & Empoyment, Endeavour Fund
Role:	Key Researcher
Total Budget:	$13.6 million
Period:	2024-2029
My Funding:	0.05 FTE per annum ($54,000)

My Contribution: Investigating how climate change affects the sea ice conditions in the Ross Sea using the P-SKRIPS model.

Program Title: Grounding zone estuaries: How can ocean stratification, baroclinicity, and tides beneath 500 m of Antarctic ice influence global sea level rise?
Funding Agency: Royal Society Te Apārangi, Te Pūtea Rangahau a Marsden | Marsden Fund
Role: Modelling Lead, Postdoctoral Fellow
Period: 2024-2027
My Funding: 0.25 FTE per annum ($120,000)
My Contribution: Investigating the Ross Ice Shelf cavity conditions with the P-SKRIPS model (MITgcm component); Supervision of PhD students.

Program Title: Extreme wildfire: Our new reality - are we ready?
Funding Agency: Hīkina Whakatutuki | Ministry of Business, Innovation & Empoyment, Endeavour Fund
Role: Key Researcher
Total Budget: $11.25 million
Period: 2021-2026
My Funding: 0.5 FTE per annum, 2023-2027 ($262,000)
My Contribution: Developed Weather-FDS: the first atmosphere-fire coupled model that is able to simulate coupled fire-atmosphere interactions

Past Grants

Program Title: Future of Antarctic Sea Ice Depends on Air-Sea Interactions
Funding Agency: University of Canterbury Faculty of Science New Idea Seeding Grant
Role: PI
Total Budget: $20,000
Period: 2024
My Contribution: Funded two Research Assistants to contribute to data analysis and Antarctic Energy Balance Station development; Supplied instrumentation for the Energy Balance Station.

Program Title: Antarctic Science Platform - The National Modelling Hub
Funding Agency: Hīkina Whakatutuki | Ministry of Business Innovation & Employment, Strategic Science Investment Fund
Role: Postdoctoral Researcher
Total Budget: $49 million
Period: 2018-2025
My Funding: 1 FTE per annum, 2020-2023 ($500,000)
My Contribution: Developed the P-SKRIPS model for the Ross Sea: the first coupled atmosphere-sea ice-ocean -ice shelf model for that region.

Publications

Peer-Reviewed Journal Articles

[1] Liangjun Yan, Zhaomin Wang, Craig Stevens, **Malyarenko, Alena**, Chengyan Liu, Denise Fernandez, and Dongxiao Wang. "Interannual Salinity Variability on the Ross Sea Continental Shelf in a Regional Ocean–Sea Ice–Ice Shelf Model. Part I: Internal Oceanic Processes". In: *Journal of Physical Oceanography* 2026 (2026).

[2] Liangjun Yan, Zhaomin Wang, Craig Stevens, **Malyarenko, Alena**, Chengyan Liu, Denise Fernandez, and Dongxiao Wang. *Interannual Salinity Variability on the Ross Sea Continental Shelf in a Regional Ocean–Sea Ice–Ice Shelf Model. Part II: Surface–Driven Processes*. 2026.

[3] D. Fernandez, C. Stevens, C. Riesselman, M. Bowen, C. Stewart, O. Truax, R. McKay, E. Behrens, L. Cornelissen, G. Cortese, M. Duffy, B. Duncan, L. Fontanot, J. Frazer, A. Gossart, S. Halfter, J. Hillman, S. Jendersie, **Malyarenko, A.**, H. Macdonald, J. McInerney, G. O'Connor, J. Prebble, N. Robinson, S. Seabrook, A.-M. Wieczorek, Y. Xiahou, L. Yan, I. Hawes, and N. Bertler. "Spatiotemporal connections in the Ross Sea: A synthesis from the New Zealand Antarctic Science Platform Ocean Mechanics project". In: *Elementa: Science of the Anthropocene* 13.1 (2025), p. 00097. ISSN: 2325-1026. DOI: 10.1525/elementa.2024.00097.

[4] Nicholas R. Golledge, Elizabeth D. Keller, Alexandra Gossart, **Malyarenko, Alena**, Angela Bahamondes-Dominguez, Mario Krapp, Stefan Jendersie, Daniel P. Lowry, Alanna Alevropoulos-Borrill, and Dirk Notz. "Antarctic coastal polynyas in the global climate system". In: *Nature Reviews Earth & Environment* (2025), pp. 1–14. DOI: 10.1038/s43017-024-00634-x.

[5] Dongqi Lin, Marwan Katurji, Peyman Zawar-Reza, **Malyarenko, Alena**, Andres Valencia, and Jiawei Zhang. "High-Fidelity Experimental Simulations of Atmospheric Flow Near Low-Intensity Fires in a Wildland-Urban Interface". In: *Journal of Geophysical Research: Atmospheres* 130.19 (2025). e2025JD043641 2025JD043641, e2025JD043641. DOI: https://doi.org/10.1029/2025JD043641. URL: https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2025JD043641.

[6] Francesca Baldacchino, Nicholas R. Golledge, Mathieu Morlighem, Huw Horgan, Alanna V. Alevropoulos-Borrill, **Malyarenko, Alena**, Alexandra Gossart, Daniel P. Lowry, and Laurine van Haastrecht. "Modelling GNSS-observed seasonal velocity changes of the Ross Ice Shelf, Antarctica, using the Ice-sheet and Sea-level System Model (ISSM)". In: *The Cryosphere* 19.1 (2024), pp. 107–127. DOI: 10.5194/tc-19-107-2025.

[7] Luke G. Bennetts, Callum J. Shakespeare, Catherine A. Vreugdenhil, Annie Foppert, Bishakhdatta Gayen, Amelie Meyer, Adele K. Morrison, Laurie Padman, Helen E. Phillips, Craig L. Stevens, Alessandro Toffoli, Navid C. Constantinou, Jesse M. Cusack, Ajitha Cyriac, Edward W. Doddridge, Matthew H. England, D. Gwyn Evans, Petra Heil, Andrew McC. Hogg, Ryan M. Holmes, Wilma G. C. Huneke, Nicole L. Jones, Shane R. Keating, Andrew E. Kiss, Noa Kraitzman, **Malyarenko, Alena**, Craig D. McConnochie, Alberto Meucci, Fabien Montiel, Julia Neme, Maxim Nikurashin, Ramkrushnbhai S. Patel, Jen-Ping Peng, Matthew Rayson, Madelaine G. Rosevear, Taimoor Sohail, Paul Spence, and Geoffrey J. Stanley. "Closing the Loops on Southern Ocean Dynamics: From the Circumpolar Current to Ice Shelves and From Bottom Mixing to Surface Waves". In: *Reviews of Geophysics* 62.3 (2024). ISSN: 8755-1209. DOI: 10.1029/2022rg000781.

[8] Yoshihiro Nakayama, **Malyarenko, Alena**, Hong Zhang, Ou Wang, Matthis Auger, Yafei Nie, Ian Fenty, Matthew Mazloff, Armin Köhl, and Dimitris Menemenlis. "Evaluation of MITgcm-based ocean reanalyses for the Southern Ocean". In: *Geoscientific Model Development* 17.23 (2024), pp. 8613–8638. DOI: 10.5194/gmd-17-8613-2024.

[9] **Malyarenko, Alena**, Alexandra Gossart, Rui Sun, and Mario Krapp. "Conservation of heat and mass in P-SKRIPS version 1: the coupled atmosphere–ice–ocean model of the Ross Sea". In: *Geoscientific Model Development* 16.11 (2023), pp. 3355–3373. DOI: 10.5194/gmd-16-3355-2023. URL: https://gmd.copernicus.org/articles/16/3355/2023/gmd-16-3355-2023.html.

[10] Francesca Baldacchino, Mathieu Morlighem, Nicholas R. Golledge, Huw Horgan, and **Malyarenko, Alena**. "Sensitivity of the Ross Ice Shelf to environmental and glaciological controls". In: *The Cryosphere* 16.9 (2022), pp. 3723–3738. DOI: 10.5194/tc-16-3723-2022.

[11] **Malyarenko, A.**, Andrew J. Wells, Patricia J. Langhorne, Natalie J. Robinson, Michael J.M. Williams, and Keith W. Nicholls. "A synthesis of thermodynamic ablation at ice-ocean interfaces from theory, observations and models". In: *Ocean Modelling* (2020), p. 101692. ISSN: 1463-5003. DOI: 10.1016/j.ocemod.2020.101692.

[12] **Malyarenko, A.**, N. J. Robinson, M. J. M. Williams, and P. J. Langhorne. "A Wedge Mechanism for Summer Surface Water Inflow Into the Ross Ice Shelf Cavity". In: *Journal of Geophysical Research: Oceans* 124.2 (2019), pp. 1196–1214. ISSN: 2169-9275. DOI: 10.1029/2018jc014594.

Accepted

[1] **Malyarenko, Alena**, Marwan Katurji, and Jiawei Zhang. "Weather-FDS: Development of the Coupled Weather-Fire Dynamics Simulator". In: *Advances on Forest Fire Research* 2026 (2026).

[2] Gitesh Wasson, Marwan Katurji, Peyman Zawar-Reza, Jiawei Zhang, Tara Strand, and **Malyarenko, Alena**. "Fire bursts: A new fire behaviour parameter to understand coupled atmosphere-fire interactions". In: *Advances on Forest Fire Research* 2026 (2026).

In Review

[1] Alexandra Gossart, **Malyarenko, Alena**, Liv Cornelissen, Craig Stevens, Una Miller, Christopher J. Zappa, Nancy Luca, Pasquale Castagno, and Giorgio Budillon. "Representation of polynyas in the Ross Sea coupled atmosphere–sea ice–ocean model P-SKRIPSv2". In: *EGUsphere* 2025 (2026), pp. 1–31. DOI: 10.5194/egusphere-2025-4332.

Peer-Reviewed Book Chapters

[1] **Alena Malyarenko**. "Ice-Ocean Boundary Layer—Part 2: Variability". In: *Encyclopedia of Ocean Sciences (Third Edition)*. Ed. by J. Kirk Cochran, Henry J. Bokuniewicz, and Patricia L. Yager. Third Edition. Oxford: Academic Press, 2019, pp. 166–174. ISBN: 978-0-12-813082-7. DOI: https://doi.org/10.1016/B978-0-12-409548-9.11487-3.

[2] M.G. McPhee, J.H. Morison, and **Alena Malyarenko**. "Ice-Ocean Boundary Layer. Part 1: Structure". In: *Encyclopedia of Ocean Sciences (Third Edition)*. Ed. by J. Kirk Cochran, Henry J. Bokuniewicz, and Patricia L. Yager. Third Edition. Oxford: Academic Press, 2019, pp. 155–165. ISBN: 978-0-12-813082-7. DOI: https://doi.org/10.1016/B978-0-12-409548-9.09552-X.

Datasets

[1] Alexandra Gossart and **Malyarenko, Alena**. *Dataset - Representation of polynyas in the Ross Sea coupled atmosphere–sea ice–ocean model P-SKRIPSv2*. Zenodo, Sept. 2025. DOI: 10.5281/zenodo.15778732.

[2] **Malyarenko, Alena** and Alexandra Gossart. *P-SKRIPS Version 2*. Sept. 2025. DOI: 10.5281/zenodo.15942874.

[3] Yoshihiro Nakayama and **Malyarenko, Alena**. *Analysis scripts developed for "Evaluation of MITgcm-based ocean reanalysis for the Southern Ocean"*. Aug. 2024. DOI: 10.5281/zenodo.13270783.

[4] **Malyarenko, Alena** and Alexandra Gossart. *P-SKRIPS Version 1*. Mar. 2023. DOI: 10.5281/zenodo.7739063.

[5] **Malyarenko, Alena** and Alexandra Gossart. *PSKRIPS Version 1 - Data for January Case.* Zenodo, Mar. 2023. DOI: 10.5281/zenodo.7739059.

TEACHING AND ADVISING

Courses

Course Title: PHYS330 Environmental and climate modelling, University of Canterbury
Term and Year Taught: Term 2, 2024
Number of Students: 28
Additional Course Information: Developed and implemented a 4-week Ocean Modelling block.

Course Title: GEOG310 Weather Systems, University of Canterbury
Term and Year Taught: Term 2, 2024
Number of Students: 20
Additional Course Information: Developed and implemented an invited lecture + lab block on El Niño-Southern Oscillation

Supervision of Graduate Students

Phyo Wai Thaw | Doctoral, University of Canterbury ENV790
February 2025 - February 2028
Project: Ice-Ocean Boundary Layer Modelling with Large Eddy Simulation

Samira Hassani | Doctoral, University of Canterbury ENV790
February 2024 - February 2027
Project: Drivers of Mesocyclogenesis in the Ross Sea

Liv Cornelissen | Doctoral, University of Auckland and Earth Sciences NZ (NIWA)
July 2023 - November 2026
Project: Modelling of The Terra Nova Bay Polynya in MITgcm
Currently: University of Otago, Teaching Fellow

Xingyu Wang | Doctoral, Victoria University of Wellington
July 2024 - January 2028
Project: Non-hydrostatic Processes at the Ross Ice Shelf Front

Wanqin Zhong | Doctoral, University of Canterbury and Wuhan University
May 2026 - January 2027
Project: Where CO_2 is absorbed, where C_{ANT} is exported, and how are these two processes linked by atmosphere-sea ice–ocean interaction?

Advising Activities

Liangjun Yan | China Scholarship Council Doctoral student, 2023
Project: Ross Sea modelling in MITgcm
Currently: Postdoctoral Researcher at Sun Yat-sen University (Guangzhou, China)

Barney Evans | Summer Project, November 2025 - April 2026
Project: Investigation of Dense Water Production in the Ross Sea in CMIP models
Currently: University of Cambridge, MPhil in Quantitative Climate and Environmental Science

Bethany Millar | RA (Faculty of Science New Idea Seeding Grant), January 2025
Project: Identification of Mesocyclones in the observational datasets

Eva Nielsen | RA (Faculty of Science New Idea Seeding Grant), October 2024
Project: Development of AWS deployment at the Inexpressible Island, Antarctica
Currently: Geological Survey of Denmark and Greenland

Educational Outreach

Course Title: **Fluids Hackathon**
Year Taught: 2023-2024
Number of Students: 10
Additional Course Information: General modelling principles, good practices in Python and LaTeX, working with satellite data in Google Collab.

Course Title: **MITgcm Modelling Course**
Year Taught: 2024
Number of Students: 5 Ocean Modellers at NIWA
Additional Course Information: Installed models, created new domains, and made test runs.

Course Title: Antarctic Science Platform National Modelling Hub **Winter School on Climate Modelling**
Year Taught: 2022, 2023, 2025
Number of Students: 20 students each year
Additional Course Information: Students with little to no modelling experiments, including postgrads, industry and faculty members, were taught to make a small ensemble of climate model output, analyse the results and create spatial plots in Python.

SERVICE & PROFESSIONAL DEVELOPMENT

Service to the Discipline

I have organised multiple conferences over the last few years in the field of Physical Oceanography. Based on that, I have led the work with the Scientific Committee on Antarctic Research SCAR EDI Action Group (https://scar.org/science/excom/edi) on recommendations on "How to organise the conference for dummies" aimed at postdoctoral researchers who often organise workshops for 50-75 people.

Physical Oceanography Workshop (POW) 2026:

- September 2026
- Currently organising a 2026 meeting of the biannual Physical Oceanography community
- Organised a 2-day program for the workshop, including a discussion session focused on building a strong Physical Oceanography Community in NZ

Forum for the Ice Shelf Processes (FRISP) - Southern Hemisphere 2025:

- https://frisp.w.uib.no/2024/10/21/frisp-south-7-10-july-2025/
- Initiated a Southern Hemisphere meeting for 30 people
- Secured funds from SCAR to support ECR travel to the meeting
- Organised a 3-day program for the workshop, including an interactive session on ice shelf cavity melt rates

Joint Snow and Ice Research Group (SIRG) and New Zealand Sea Ice Symposium (NZSIS) Meeting

- Led organisation for the NZSIS part; 42 Attendees
- Program planning
- Venue Booking

Physical Oceanography Workshop (POW) 2024:

- Organised a 2024 meeting of the biannual Physical Oceanography community, 34 attendees
- Organised a 2-day program for the workshop, including a discussion session focused on building a strong Physical Oceanography Community in NZ
- Resulting in organising a Regular Physical Oceanography catch-up meeting, held monthly online

Peer Review:

- Nature Communications Earth & Environment
- Journal of Advances in Modeling Earth Systems
- Geophysical Research Letters
- Environmental Research Letters
- Journal of Geophysical Research - Oceans
- Deep-Sea Research Part II
- Frontiers in Marine Science
- Advances in Atmospheric Science
- Journal of Climate
- Journal of Oceanology and Limnology

Service to the Community and Public

Organiser of the Antarctic Family Day, 10 August 2019, Miramar Central School hall
https://www.eventfinda.co.nz/2019/antarctic-family-day/wellington

Professional Distinctions

2026 - present - SOOS Ross Sea Working Group – Leadeship team 2024 - 2026 - SOOS Ross Sea Working Group – ECR Representative

2023 - present - National Representative for FRISP (The Forum for Research into Ice Shelf Processes, SCAR)

Awards

2016 – New Zealand Antarctic Research Institute (NZARI) Scholarship, $20,000 plus logistics support from Antarctica New Zealand

2016 – University of Otago Doctoral Scholarship, course costs plus $25,000 pa for three years

Presentations

Invited Seminars

[1] A. Malyarenko. “The meltwater in the Ross Sea: where did it come from, where will it go?” University of Canterbury, SPCS Seminar. Feb. 2024.

[2] A. Malyarenko, A. Gossart, A. Pletzer, and C. Scott. “Enhancing Code Reusability: Using Containers to port an Atmosphere-Ocean Coupled Model on NeSI”. NeSI Invited Webinar. Apr. 2024.

[3] A. Malyarenko, Y. Nakayama, and C. Stevens. “The Ross Sea Has Been Freshening ... Does The Ross Ice Shelf Cavity Care About It?” Institute of Low Temperature Science, Hokkaido University — Sapporo, Japan. July 2022.

[4] A. Malyarenko, A. J. Wells, P. J. Langhorne, N. J. Robinson, and M. J. M. Williams. “Modelling of Ice melting: Theory of dissolving, practical applications to ice shelf-ocean modelling”. Hokkaido University — Hakodate, Japan. July 2022.

[5] A. Malyarenko, C. Stevens, Y. Nakayama, A. Gossart, and R. Sun. “How do you solve a problem like ... Ross Ice Shelf cavity modeling?” Joint KOPRI-NIWA seminar series POET (Online). Sept. 2021.

[6] A. Malyarenko, A. J. Wells, P. J. Langhorne, N. J. Robinson, M. J. M. Williams, and K. W. Nicholls. “Ice shelf-ocean ablation: a wide range in observations, a wider range in modelling”. The Oregon State University seminar (Online). Sept. 2020.

[7] A. Malyarenko, P. J. Langhorne, N. J. Robinson, and M. J. M. Williams. “Thermodynamic balance at ice-ocean interface in theory, models and maybe real life”. The British Antarctic Survey seminar (Cambridge, UK). Oct. 2017.

Oral Presentations

[1] A. Malyarenko, A. Gossart, A. Pauling, C. Stevens, D. Fernandez, and N. Robinson. "Saviour to Villain? Sea ice production on the Ross Sea continental shelf will promote transition into a warm cavity by the end of the century." Forum for Research into Ice Shelf Processes (FRISP) workshop — Oslo, Norway. Aug. 2026.

[2] A. Malyarenko, A. Gossart, A. Pauling, C. Stevens, D. Fernandez, and N. Robinson. "Saviour to Villain? Sea ice production on the Ross Sea continental shelf will promote transition into a warm cavity by the end of the century." SCAR Open Science Conference — Oslo, Norway. Aug. 2026.

[3] A. Malyarenko, A. Gossart, and Y. Nakayama. "Is it possible to model the future of the RIS cavity with the present day tools?" Physical Oceanography Workshop — Wellington, New Zealand. Sept. 2024.

[4] A. Malyarenko, A. Gossart, and Y. Nakayama. "The Ross Sea in the new coupled model Polar-SKRIPSv2". SCAR Open Science Conference — Pucón, Chile. Aug. 2024.

[5] A. Malyarenko, A. Gossart, and Y. Nakayama. "The Terra Nova Bay Polynya activity in the new coupled model Polar-SKRIPSv1". The Southern Ocean Observing System (SOOS) Symposium — Hobart, Australia. Aug. 2023.

[6] A. Malyarenko, A. Gossart, and Y. Nakayama. "The Terra Nova Bay Polynya activity in the new coupled model Polar-SKRIPSv1". Antarctica New Zealand Latitudes of Change conference — Christchurch, New Zealand. July 2023.

[7] A. Malyarenko, A. Gossart, and Y. Nakayama. "The Terra Nova Bay Polynya activity in the new coupled model Polar-SKRIPSv1". Japan Geoscience Union Meeting — Tokyo, Japan. May 2023.

[8] A. Malyarenko and A. Gossart. "Antarctic polynyas in a coupled model". Physical Oceanography Workshop — Wellington, New Zealand. 2022.

[9] A. Malyarenko, A. Gossart, and R. Sun. "Balancing fluxes through WRF - MITgcm interface in the Scripps-KAUST model for the Ross Sea Region". 13th International Conference on Southern Hemisphere Meteorology and Oceanography (ISCHMO) — Online. Feb. 2022.

[10] A. Malyarenko, Y. Nakayama, C. Stevens, S. Jendersie, and M. Krapp. "The Ross Sea Has Been Freshening ... Does The Ross Ice Shelf Cavity Care About It?" 2022 Ocean Sciences Meeting — Online. Feb. 2022.

[11] A. Malyarenko and A. Gossart. "Coupled Climate-Ice-Ocean Modelling: Setting up a framework". New Zealand Research Software Engineering Conference 2021 (NZRSE) — Online. Sept. 2021.

[12] A. Malyarenko, S. Jendersie, M. J. M. Williams, N. J. Robinson, P. J. Langhorne, C. Stevens, and D. Gwyther. "Vertical mixing approaches to the ice shelf-ocean boundary layer". Physical Oceanography Workshop (POW) — Online. Nov. 2020.

[13] A. Malyarenko, P. J. Langhorne, M. J. M. Williams, and N. J. Robinson. "Phase change and the ocean boundary layer". Sea Ice Symposium — Wellington, New Zealand. July 2018. Student presentation runner-up.

[14] A. Malyarenko, M. J. M. Williams, N. J. Robinson, and P. J. Langhorne. "Inflow of Summer Surface Water into the Ross Ice Shelf Cavity". Forum for Research into Ice Shelf Processes (FRISP) workshop (32nd) — Aussois, France. Sept. 2018.

[15] A. Malyarenko, P. J. Langhorne, M. J. M. Williams, and N. J. Robinson. "Interaction between phase change and boundary layer structure". Workshop at the Isaac Newton Institute, "Ice-fluid interaction" — Cambridge, UK. Oct. 2017.

[16] A. Malyarenko, P. J. Langhorne, M. J. M. Williams, and N. J. Robinson. "Is the Ross Ice Shelf Melting or Dissolving, and Why Does It Matter?" University of Otago Student Research Symposium — Dunedin, New Zealand. Aug. 2017.

[17] A. Malyarenko, P. J. Langhorne, M. J. M. Williams, and N. J. Robinson. "Is the Ross Ice Shelf melting or dissolving, and why does this matters?" New Zealand Antarctic Science Conference — Dunedin, New Zealand. June 2017.

[18] A. Malyarenko, P. J. Langhorne, M. J. M. Williams, and N. J. Robinson. "Is the Ross Ice Shelf Melting or Dissolving?" Physical Oceanography Workshop — Wellington, New Zealand. Aug. 2017.

Poster Presentations

[1] A. Malyarenko, A. Gossart, and A. Pauling. "The Ross Ice Shelf cavity in the new coupled model Polar-SKRIPSv2". Climate and Cryosphere Open Science Conference — Wellington, New Zealand. Feb. 2026.

[2] A. Malyarenko, A. Gossart, and A. Pauling. “Ice shelf - ocean interaction in the Ross Sea - how much will it change?” Ocean Modelling and Observations Workshop — Canberra, Australia. July 2024.

[3] A. Malyarenko, S. Jendersie, M. J. M. Williams, N. J. Robinson, and P. J. Langhorne. “Effects of model resolution on ice shelf-ocean boundary layer”. EGU Virtual Conference. May 2020.

[4] A. Malyarenko, S. Jendersie, M. J. M. Williams, N. J. Robinson, P. J. Langhorne, C. Stevens, and D. Gwyther. “Vertical Mixing Approaches to the Ice Shelf-Ocean Boundary Layer”. AGU Virtual Conference. Dec. 2020.

[5] A. Malyarenko, S. Jendersie, M. J. M. Williams, N. J. Robinson, P. J. Langhorne, C. Stevens, and D. Gwyther. “Vertical Mixing Approaches to the Ice Shelf-Ocean Boundary Layer”. Geosciences Conference — Christchurch, New Zealand. Nov. 2020.

[6] A. Malyarenko, A. J. Wells, P. J. Langhorne, N. J. Robinson, M. J. M. Williams, and K. W. Nicholls. “Ice shelf ablation and ocean boundary layer interaction”. SCAR Virtual Conference. Aug. 2020.

[7] A. Malyarenko, P. J. Langhorne, N. J. Robinson, and M. J. M. Williams. “How to model a dissolving Ross Ice Shelf?” Workshop in the Isaac Newton Institute — Cambridge, UK. Sept. 2017.

[8] A. Malyarenko, M. J. M. Williams, N. J. Robinson, and P. J. Langhorne. “Ocean and ice shelf interaction in the Southwest Ross Sea”. The International Symposium on the Cryosphere in a Changing Climate — Wellington, New Zealand. Feb. 2017.